HW 3

Due on 20 February 2026 at 8PM on Gradescope.

This assignment is worth 25 points, and you should be spending roughly 10 minutes per point on average. If you find yourself spending more than $10n$ minutes on a n-point problem, reach out to me via drop-in hours, Discord, or email.

Collapse all exercises

Goals

- Group theory intuition
- Modular arithmetic
- More permutations and symmetry groups
- Introductory proofs about groups

▼ EXERCISE 1 [2]

Either save this exercise for last, or do it incrementally as you work on the homework. You may need to add `\usepackage{xcolor}` to the preamble of your document.

1. Once you have completed the assignment, fill out the following template listing collaborators, tools, and a reflection on each problem you worked on.

```
Collaborators: Alex, Ross, Margaret, Joan

\noindent
Tools and resources used: On Exercise 1, I used ChatGPT to help
\begin{enumerate}
  \item[Exercise 1]
    {\color{red} Points attempted: 7}.
    {\color{blue} Time spent: 56 minutes}.
    {\color{brown} Difficulty: 3/10}.
    {\color{green!50!black} Self-assessed score: 6/7}.
    {\color{magenta} Questions/Feedback: None.}
  \item[Choice Exercise 7]
    {\color{red} Points attempted: 3}.
    {\color{blue} Time spent: 18 minutes}.
    {\color{brown} Difficulty: 7/10}.
    {\color{green!50!black} Self-assessed score: 3/3}.
    {\color{magenta} Questions/Feedback: How should I be thinki
\end{enumerate}
```

2. Choose one or two problems that you would most like feedback on, with a brief note of what you would most like me to focus on. (This could be "mathematical correctness," "grammar," "notation," "formatting," or anything else.)

▼ EXERCISE 2 [2]

Post on the `#hw-prompt` Discord channel with a screenshot of a homework write-up from this current homework or a previous homework that you feel proud of or where you had an "aha" moment while working on it. Make sure to post the number of the problem, remind us of the context of what the problem is about, and write at least one thought that you have about the problem or your write-up.

▼ EXERCISE 3 [6]

1. Let $(G, *) = (\mathbb{Z}, +)$, and let $a = 1 \in G$ and $b = 3 \in G$. Compute $a^4 * b^{-2} \in \mathbb{Z}$.
(Notice that superscripts aren't doing the same thing that they do in ordinary arithmetic!)
2. Let $g \in G$, and let $\langle g \rangle$ be the cyclic subgroup of G generated by g. Prove that the order of the element g is equal to the order of the subgroup $\langle g \rangle$.
3. Consider the cyclic group C_{20}, where r is an 18° rotation, and thus $|r| = 20$. Find all elements $r^n \in C_{20}$ such that $\langle r^n \rangle = C_{20}$.
4. Give an example of an infinite group $(G, *)$ and a subset $H \subseteq G$ which is closed under $*$, but where H is not a subgroup of G.
5. Prove that if $a \equiv a' \pmod{n}$, $b \equiv b' \pmod{n}$, and $a + b \equiv c \pmod{n}$, then

$$a' + b' \equiv c \pmod{n}.$$

Choice Exercises

Do 15 points of the following.

▼ CHOICE EXERCISE 4 [≤14] (REVIEW/PRACTICE)

1. [2] Certain mattress manufacturers recommend flipping a mattress every 6 months to prevent uneven wear, alternating between each of its four symmetries.

a. Prove that the "mattress flipping" group cannot be generated by a single element, which means that you cannot do the same procedure each six months if you want even wear.

b. Explain why flipping the mattress over each spring and rotating it 180° each fall will hit each of the four symmetries equally.

2. [2] Suppose G is a group with a subgroup $H \leq G$. Prove that if $h \in H$ and $g \in G \setminus H$ (i.e. g is in G but not in H) then $g * h \notin H$.

3. [2] Explain why the inverse of the permutation $(12345)(6)(789)$ is $(987)(6)(54321)$. And use this principle find the inverse of $\pi = (726314)(95)$ in cycle notation.

4. [1] Give an example of three permutations $\pi_1, \pi_2, \pi_3 \in S_9$ that commute with $\sigma = (125)(79)$, and two that don't.

5. [2] There are two elements in D_4 that commute with every other element. In other words, there exist two distinct elements g_1 and g_2 such that $a * g_1 = g_1 * a$ and $a * g_2 = g_2 * a$ for all $a \in D_4$. Find them.

6. [2] Prove that if G is a finite group of order $n > 2$, then there are no subgroups of G of order $n - 1$.

7. [1] Prove that the subgroup relation is transitive: if $H \leq G$ and $K \leq H$, then $K \leq G$.

8. [1] Give an example of a group G and two elements $a, b \in G$ such that $(a * b)^2 \neq a^2 * b^2$.

9. [1] Find the smallest nonnegative integer k such that $1234 \equiv k \pmod{11}$.

▼ Choice Exercise 5 [≤9] (proofs)

1. [2] Replace each instance of $\sim$ in the proof environment in order to fill in a proof of the following theorem. (Recall you can copy/paste the math on the website to get LaTeX.)

Theorem. If $(G, \star)$ is a finite group, and $g \in G$, then the order of g is finite.

Proof. Recall the definition of the order of an element: ~ ~ ~ ~ ~.
Suppose that $|G| = n$. Now consider the sequence of elements $(g^1, g^2, g^3, \dots, g^{n+1})$. There must exist some g^k, g^m (with $k < m$) such that $g^k = g^m$ because ~ ~ ~ ~ ~.
Thus, we can multiply both sides by

$$
\begin{aligned}
g^k &= g^m \\
g^k \star (\ ) &= g^m \star (\ ) \\
(\ \ ) &= (\ \ )
\end{aligned}
$$

Therefore $g \ = e \in G$, so ~ ~ ~ ~.

2. [4] Let $a \in (G, *)$ and recall that we define $\langle a \rangle$ as $\langle a \rangle = \{a^n \mid n \in \mathbb{Z}\}$. Recall that we define

$$a^n = \underbrace{a * a * \cdots * a}_{n \text{ times}} \qquad \text{and} \qquad a^{-n} = \underbrace{a^{-1} * a^{-1} * \cdots * a^{-1}}_{n \text{ times}}.$$

We will prove that $a^k * a^m = a^{k+m}$ for all $k, m \in \mathbb{Z}$.

a. Start by explaining why if $k \geq 0$ and $m \geq 0$, then $a^k * a^m = a^{k+m}$.

b. Explain why an essentially similar argument shows that if $k \leq 0$ and $m \leq 0$, then $a^k * a^m = a^{k+m}$.

c. Now, let $n = -k$.

i. Show that if $m > k$ then $a^m * (a^{-1})^k = a^{m+n}$.

ii. Show that if $m = k$ then $a^m * (a^{-1})^k = a^{m+n}$.

iii. Show that if $m < k$ then $a^m * (a^{-1})^k = a^{m+n}$.

d. State why the above is done "without loss of generality" with respect to the negative exponent being on the right. In other words, write a sentence stating why the proofs where the inverses are on the left are essentially the same.

e. An abelian group is a group where $g_1 * g_2 = g_2 * g_1$ for all $g_1, g_2 \in G$. Use the identity $a^k * a^m = a^{k+m}$ to prove that $\langle a \rangle$ is an abelian group.

3. [2] Let $(G, *)$ be a group and let $a \in G$. Let H be the set of elements that commutes with a:

$$H = \{g \in G \mid g * a = a * g\}.$$

Prove or disprove whether H is a subgroup of G. (Sometimes H is denoted $C_G(\{a\})$.)

4. [1] Prove or disprove. If A and B are $n \times n$ matrices, and $AB = A$, then B is the identity matrix, $B = I_n$.

▼ CHOICE EXERCISE 6 [≤15] (PROGRAMMING)

1. [4] Prove that the set of 3×3 matrices that have one ± 1 per row and column forms a group under matrix multiplication. This group is called the octahedral group, sometimes written B_3. Without looking it up, determine the order of this group.

2. [2] Download OpenSCAD, write a `module` named `blob` that makes something asymmetrical. You can use any tools that you would like to learn how to do this, but just say which tools you used. Here's my blob, which I display by typing it below the module definition.

```
module blob() {
  cylinder(20,10,5);
  translate([-5,0,10]) rotate([45,0,0]) cube([20,10,5]);
}

blob();
```

3. [4] Apply all possible matrices from part 1 to this using `multmatrix(...)`. (You probably want to write a program that prints OpenSCAD code instead of doing this by hand.)

```
color("red")
multmatrix([
  [0, -1, 0],
  [0,  0, 1],
  [1,  0, 0]
]) { blob(); }
```

4. [1] Do this again, but color the parts whose corresponding matrix has determinant 1 different from the color of those whose corresponding matrix has determinant -1.

5. [1] Post on the `#hw-prompt` channel on Discord with a screenshot of `blob`, a screenshot of all of the rotations of `blob`, and an STL export of both.

6. [3] 3D print the object (perhaps by using the Maker Studio), and post an image to Discord.

▼ Choice Exercise 7 [2 + 2] (video/self-study)

1. Watch Ryota Matsuura's YouTube video "Chapter 4: Modular Arithmetic" through "Problem #7," which is first shown at timestamp 10:32. Do the problem,

and write-up your solution here.

Chapter 4: Modular Arithmetic

Ryota Matsuura

Watch on

2. Set a timer for 20 minutes, and read whatever you want from the Judson book—I recommend glancing at Chapters 3.1, 4.2, and 6.1, but there is a lot of good stuff in there. Write 2--3+ sentences about what you liked, didn't like, understood, felt confused about, felt curious about, or anything else.

▼ Choice Exercise 8 [4] ⭐

Here we will study the subgroups of the symmetries of the square,

$$D_4 = \{e, r, r^2, r^3, f, rf, r^2f, r^3f\}$$

1. Write down the (trivial) subgroup of order 1 and the (improper) subgroup of order 8, and say what these mean geometrically.

2. Write down the five subgroups of order 2, and describe what they mean geometrically.

3. Write down the three subgroups of order 4, and describe in words what they mean geometrically.

4. What do you notice about the orders of the subgroups?

5. Are any of these subgroups "essentially similar" to each other? If so, which ones?

▼ Final Choice Exercise [≤7]

Do any subset of the following parts, state which you did and how many points you are claiming for this problem (maximum of 7 points).

1. [≤3] Email Prof Kagey (pkagey@cpp.edu) or post in the `#feedback` channel on Discord with a typo that you have spotted in this assignment or with a

suggestion about how to make something clearer. (One point for every six minutes that the typo got you stuck, with a maximum of three points.)

2. [≤2] Ask a question on Discord. (One point per question, up to 2.)

3. [≤4] Answer a classmate's question on Discord or give a hint about a problem on this week's assignment.[1] (Two points per answer/hint, up to 4.)

4. [≤4] Can you generalize any of the above problems in an interesting way? If so say what the generalization is, attempt to solve it, say something about it, and say how many points you feel like your effort was worth, up to 4.

5. [≤4] Choose one or more problems from our textbook or a previous homework assignment which interest you and write the problem statement along with an answer. Say how many points you feel like your effort was worth, up to 4.

6. [2] Stop by student drop-in hours.

7. [2 or 5] Work on at least one of the above exercises with a classmate. State which classmate you worked with and which exercises you collaborated on. If you've collaborated with someone for the first time, you get 5 points, if you've collaborated with someone before, you get 2 points for working with them again.

8. [2] Turn in the assignment on time (without requesting an extension).

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[1] The syntax for a hint on Discord is double bars `||` on each side of the spoiler like this: `Remember that ||1 + x < x + 2||.` ↩

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